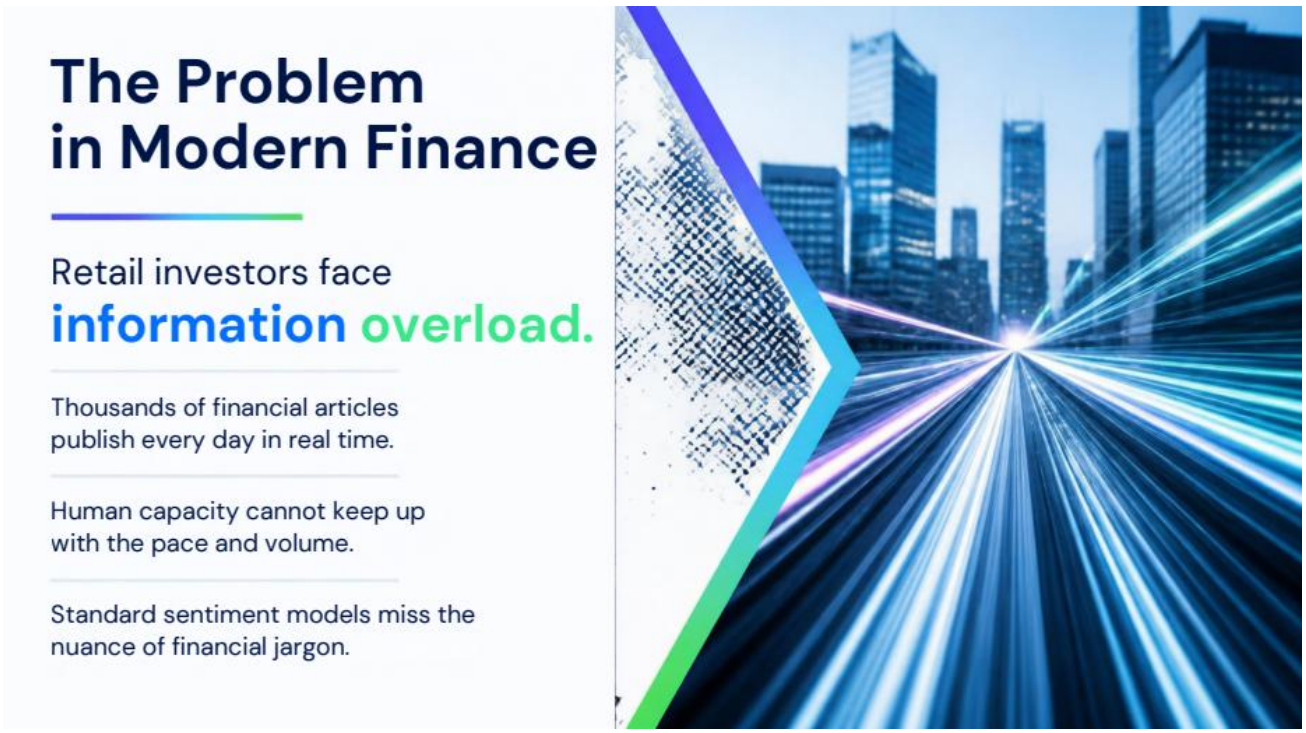
Experiment No. 10

Aim : Implementation and Presentation of Mini Project.

Project Title : FinSent – Financial Sentiment Analysis (Real-time Market Trend Prediction using NLP & Large Language Models)





Project Objective

Develop an automated NLP pipeline to instantly classify market news.

Use Generative AI to extract the single most actionable highlight from articles.

Autonomously generate AI-driven SWOT analyses and Peer Comparisons.



Resources & Technology Stack

Backend	Python, FastAPI, yfinance for real-time market data.
NLP Engine	OpenRouter API utilizing the Google Gemma LLM.
Frontend	Vanilla JavaScript, HTML5, and advanced CSS Glassmorphism.
Visualization	Chart.js for rendering dynamic technical price/volume graphs.
Data Source	yfinance for real-time market data.



Algorithm & Processing Logic

01

User inputs a stock ticker symbol.

02

Backend fetched live historical data and breaking news headlines.

03

A strict prompt is sent to the LLM to process sentiment, extract highlights, and build a SWOT matrix.

04

The LLM returns structured getJSON data.

05

Structured data is dynamically rendered onto the web dashboard.

Innovative Project Features

01

Context-Aware Styling

News highlight boxes dynamically change color based on AI sentiment.

02

Dual-Theme UI

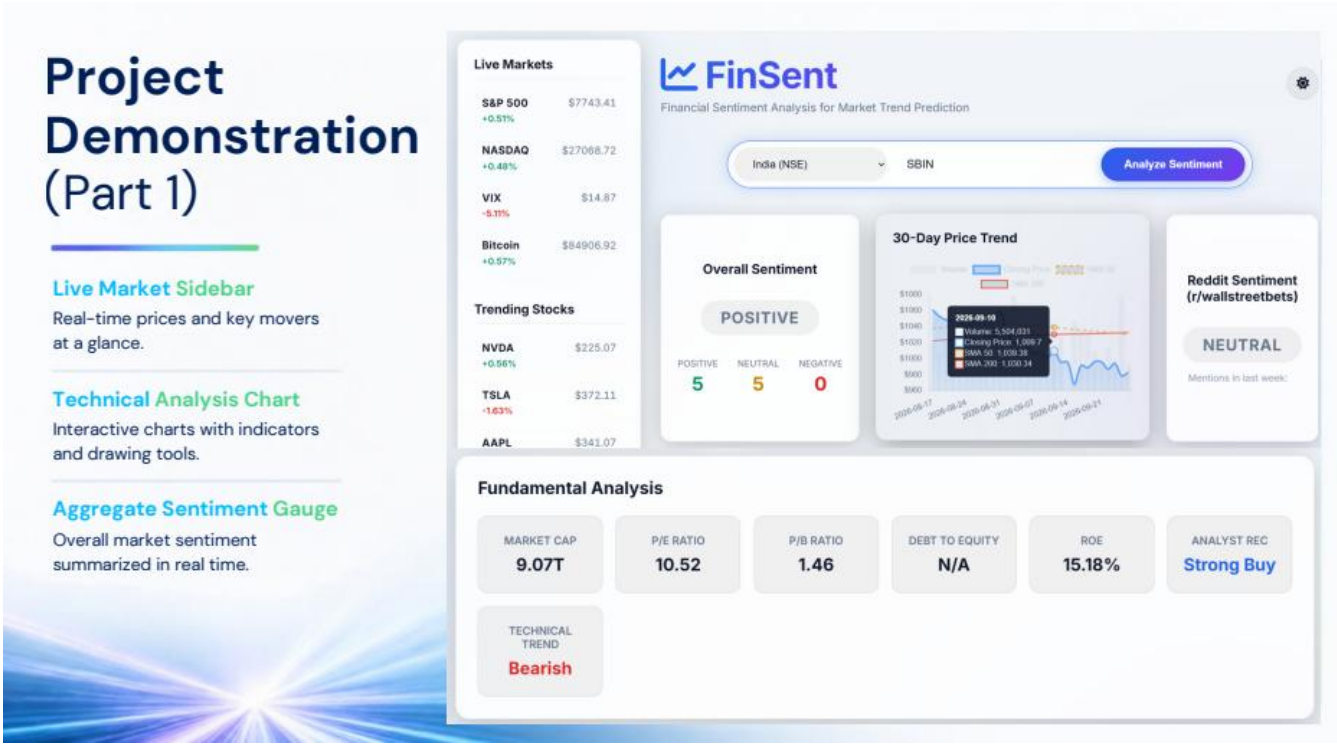
Institutional-grade Pitch Black dark mode and Light mode.

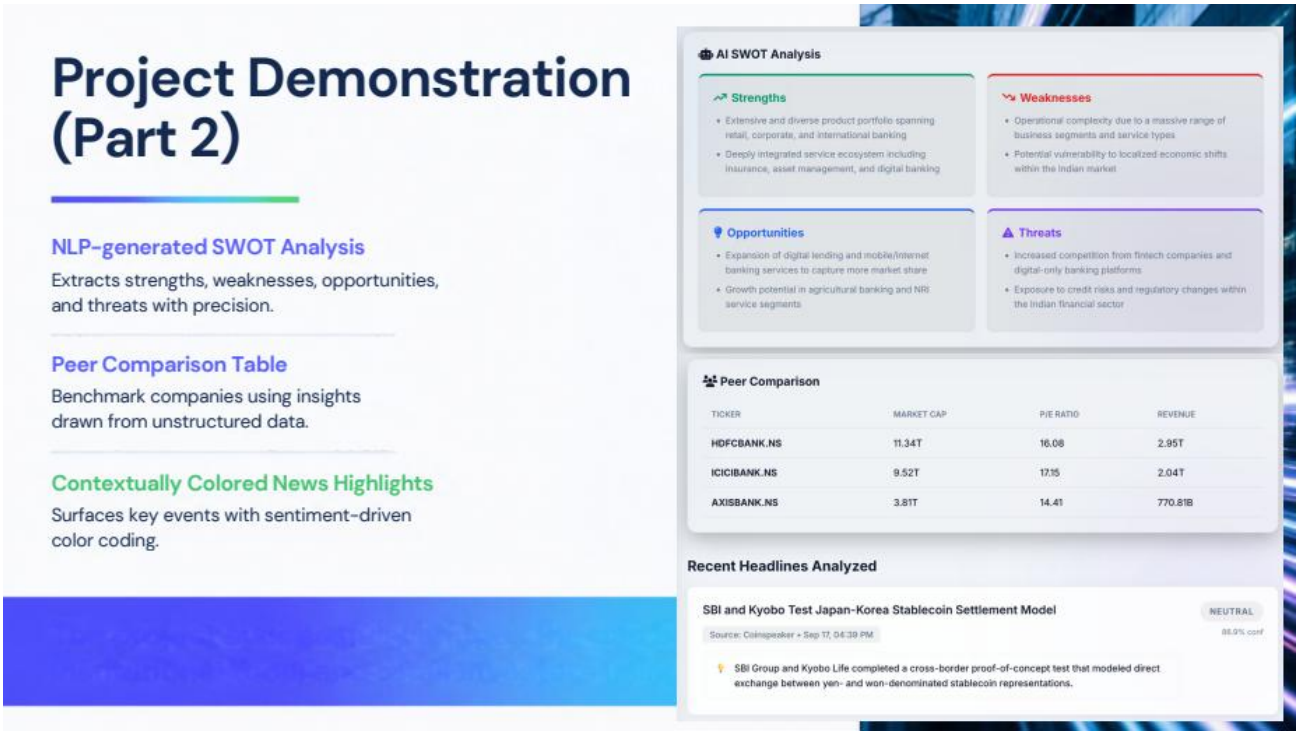
03

Automated Fundamentals

Instantly calculates Market Cap, P/E ratios, and SMA trendlines.







The project successfully bridges the gap between institutional tools and retail investors using applied NLP.

Thank You...

Conclusion : Hence, we have successfully completed our project and presentation for NLP.